

Solar Rooftop PV | Techno-Commercial Case Study

5 kWp-class Grid-Connected Rooftop Solar System | Prepared for GET (Technical Sales, Business Development & Product Management) interview preparation

This document mirrors the accompanying Excel workbook. It is a self-directed mock techno-commercial case study using public reference prices and manufacturer datasheets - not a live customer engagement or vendor quotation. All assumptions are labelled explicitly throughout.

Final Summary

Technical Configuration

Recommended configuration	9 x 575 Wp Waaree bifacial modules + Loom Solar Fusion 63 (6 kW, 3-phase)
DC system size	5.175 kWp
Estimated annual generation, Year 1	6,800 kWh
Generation yield check	1,314 kWh/kWp/year

Commercial Summary

Total direct project cost	Rs. 2,48,526
Gross project price, before subsidy	Rs. 2,78,349
PM Surya Ghar CFA applied	Rs. 78,000
Net customer investment, after subsidy	Rs. 2,00,349

Customer Economics

Net annual savings, Year 1	Rs. 47,310
Simple payback	4.2 years
Approx. 25-year ROI on net investment	~530%

Inverter Selection Note

A Loom Solar Fusion 63 (6 kW, three-phase) was selected for this 5.175 kWp array, giving a DC/AC ratio of ~0.86 (below the typical 1.1-1.25 EPC target). This is a sizing trade-off made under the assumption of a three-phase customer connection and design headroom - see Component Selection for the full rationale and the single-phase alternative that was considered.

Assumptions

Customer Profile (simulated - clearly an assumption, not real data)

Item	Value	Basis / Note
Customer type	Urban residential	Homeowner with independent rooftop access
Location	Delhi-NCR (representative)	Chosen for consistency with Vista's Noida base
Avg. monthly electricity bill	Rs. 5,500	Used to size the system to a 'higher-consumption home' segment
Assumed electricity tariff	Rs. 8/kWh	Mid-slab residential tariff typical for a Delhi-NCR high-consumption household
Monthly consumption (derived)	687.5 kWh	Bill / tariff
Usable shadow-free roof area	550-600 sq ft	Fits a ~5 kWp array at typical module footprint
Customer objective	Reduce electricity bill; moderate upfront cost priority over max ROI	Drives system sizing and component choice
Electrical connection	Three-phase	Required for the selected inverter (Fusion 63)

Technical / Generation Assumptions

Item	Value	Basis / Note
Peak sun hours (Delhi-NCR)	4.5 hrs/day	Conservative planning value for the region
System performance ratio (PR)	80%	Module/inverter/temperature/wiring/soiling losses combined
Module degradation (annual, after yr 1)	0.55%/yr	Reflects Waaree's linear performance warranty profile
O&M cost (annual)	Rs. 2,500	Cleaning + minor upkeep
Inverter replacement	Assumed once, Year 12	String inverters typically warrantied 5 yrs, useful life 10-15 yrs
Self-consumption share of generation	55%	Rest exported under net metering
Net metering credit rate	Rs. 6.5/kWh	Typically at or below retail tariff under most DISCOM rules

Government Policy (fact, not assumption)

Item	Value	Status
PM Surya Ghar CFA - 1st kW	Rs. 30,000/kW	Fact - official MNRE slab
PM Surya Ghar CFA - 2nd kW	Rs. 30,000/kW	Fact - official MNRE slab
PM Surya Ghar CFA - 3rd kW	Rs. 18,000/kW	Fact - official MNRE slab
Max residential CFA (systems ≥ 3kW)	Rs. 78,000 flat cap	Fact - per MNRE guidelines
Domestic content requirement	Yes	Waaree is domestic, so this system qualifies

Load & Requirement Analysis / System Sizing

Step 1: Estimate load from the customer's bill

Avg. monthly bill	Rs. 5,500
Assumed tariff	Rs. 8/kWh
Estimated monthly consumption	687.5 kWh
Estimated annual consumption	8,250 kWh

Step 2: Size to target ~85% of annual consumption offset

Target offset of annual consumption	85%
Target annual generation	7,012.5 kWh
Peak sun hours	4.5 hrs/day
Performance ratio	80%
Indicative required capacity	~5.34 kWp

Step 3: Fit to available module count and roof area

Selected module rating	575 Wp
Number of modules selected	9
Resulting DC system size	5.175 kWp
Check vs. indicative requirement	Within 15% - acceptable fit

Why 5.175 kWp: the customer's bill (Rs. 5,500/month at Rs. 8/kWh) implies roughly 8,250 kWh/year of consumption. Targeting 85% offset needs roughly 5.3 kWp under Delhi-NCR irradiation. Nine 575 Wp modules give 5.175 kWp DC - the closest whole-module fit that also comfortably suits the assumed roof area.

Component Selection

Module: Waaree BIN-08-575 (575 Wp, N-Type TOPCon, Bifacial, Dual-Glass)

Rated power	575 Wp
Technology	N-Type TOPCon, bifacial, dual-glass, monocrystalline
Cell count	144
Rated efficiency	22.26%
Voc (open-circuit voltage)	52.3 V
Isc (short-circuit current)	13.87 A
Dimensions	~2278 x 1134 x 33 mm
Warranty	12-yr product / 30-yr linear power output

Why this module: Bifacial N-Type gives a modest generation uplift (typically 3-8%) from rear-side reflected light. This model conservatively does NOT add a bifacial gain to the 6,800 kWh/year generation figure, since a flush-mounted residential roof install captures little rear-side gain in practice.

Inverter Selection & Sizing Rationale

This 5.175 kWp array is paired with a Loom Solar Fusion 63 (6 kW, three-phase) inverter. A smaller, single-phase inverter (~4 kW class) was also considered, since it would give a DC/AC ratio closer to the typical 1.1-1.25 EPC target. The 6 kW three-phase unit was selected instead under the assumption that the customer has (or will arrange) a three-phase connection, and to keep headroom against a Voc-derating margin. This is a sizing trade-off, not a technical error - both configurations are electrically valid for this array; the choice depends on the customer's actual electrical connection, which should be confirmed on-site before finalizing a real quotation.

	Option	Note
Alternative considered	Single-phase ~4 kW-class inverter (e.g. Fusion 41)	DC/AC ratio closer to ~1.15-1.29, but max PV input (~4.5 kW) is close to/below the 5.175 kWp array - would need array de-rate or model confirmation
Selected inverter	Loom Solar Fusion 63 (6 kW, three-phase)	Max PV input power 7.8 kW, max input voltage 1000V, 2 MPPT trackers, 40A max current, 98.5% max efficiency
Assumption relied on	Three-phase customer connection	Reasonable for a higher-consumption home; stated explicitly as a design input

String Voltage & Current Compatibility Check

Check	Result	Note
String Voc (9 modules in series)	470.7 V	Well within Fusion 63's 1000V max input, even with cold-weather Voc increase
String Isc	13.87 A	Comfortably within Fusion 63's 40A max input current
DC/AC ratio	0.86	Below the typical 1.1-1.25 target - reflects the three-phase/headroom design choice, not a sizing error

Inverter Specs Used (Loom Solar Fusion 63)

Rated output power	6 kW
Phase	Three-phase, 400V output
Max PV input power	7.8 kW
Max PV input voltage	1000 V
MPPT trackers	2
Max efficiency	98.5%
Communication	WiFi, GPRS, RS-485, RS-232
Protection	Anti-islanding, over-voltage/current/temp, DC reverse polarity, Type II SPD

Bill of Materials & Project Costing

Major Equipment

Item (Qty)	Unit Cost	Total
Waaree 575 Wp bifacial module (9 nos)	Rs. 11,299	Rs. 1,01,691
Loom Solar Fusion 63, 6kW 3-phase (1 no)	Rs. 42,000	Rs. 42,000

Balance of System

Item	Qty/Unit	Total
Aluminium/GI mounting structure	1 lot	Rs. 25,000
4 sq.mm DC solar cable	1 lot	Rs. 7,000
AC cable + accessories	1 lot	Rs. 5,000
MC4 connectors + DC isolator	1 lot	Rs. 3,500
ACDB/DCDB + AC & DC SPD (Type II)	1 lot	Rs. 8,000
MCB/MCCB protection	1 lot	Rs. 4,000
Earthing (2 pits)	1 lot	Rs. 6,000
Lightning protection	1 lot	Rs. 3,000
External display/monitoring accessory	1 lot	Rs. 2,500

Installation & Engineering

Item	Qty/Unit	Total
Installation & commissioning labour	1 lot	Rs. 15,000
Transportation	1 lot	Rs. 3,000
Engineering/design + net-metering support	1 lot	Rs. 8,000
Testing & commissioning documentation	1 lot	Rs. 3,000

Costing Roll-up

Subtotal (materials + BOS + installation)	Rs. 2,36,691
Contingency (5%)	Rs. 11,835
TOTAL DIRECT PROJECT COST	Rs. 2,48,526
EPC overhead & commercial margin (12%)	Rs. 29,823
GROSS PROJECT PRICE (before subsidy)	Rs. 2,78,349
System size	5.175 kWp
Applicable CFA (flat cap, system ≥ 3kW)	Rs. 78,000
NET CUSTOMER INVESTMENT (after subsidy)	Rs. 2,00,349

Module and inverter costs are anchored to public retail/market references; BOS, installation and engineering costs are engineering estimates typical for a residential rooftop job of this size, not confirmed supplier quotations.

Annual Generation, Savings & Financial Analysis

Annual Generation (reconciled against the 6,800 kWh/year target)

DC system size	5.175 kWp
Peak sun hours	4.5 hrs/day
Performance ratio	80%
Estimated annual generation, Year 1	6,800 kWh
Sanity check: generation yield	1,314 kWh/kWp/year

This reconciles to ~6,800 kWh/year - the confirmed baseline, superseding an earlier draft figure of 6,570 kWh which had used 5 kWp nominal instead of the actual 5.175 kWp DC size. Typical Delhi-NCR yield range is ~1,300-1,600 kWh/kWp/year, so 1,314 sits at the conservative end - appropriate for a real install rather than a lab-ideal estimate.

Bill Savings (not 1:1 with generation)

Self-consumption share	55%
Self-consumed units	3,740 kWh/yr
Exported units	3,060 kWh/yr
Retail tariff	Rs. 8/kWh
Net-metering export credit rate	Rs. 6.5/kWh
Savings from self-consumption	Rs. 29,920/yr
Credit from grid export	Rs. 19,890/yr
O&M cost	(Rs. 2,500/yr)
NET ANNUAL SAVINGS, Year 1	Rs. 47,310

Payback & ROI

Net customer investment (after subsidy)	Rs. 2,00,349
SIMPLE PAYBACK	4.2 years

Net investment / Year-1 net annual savings. Ignores degradation and tariff escalation - a conservative simple-payback view.

25-Year Perspective

Annual module degradation	0.55%/yr
Inverter replacement cost (Year 12, one-time)	Rs. 42,000
Approx. cumulative net savings over 25 years	~Rs. 10.6 lakh
Approx. 25-year ROI (on net investment)	~530%

This model deliberately uses SIMPLE payback as the headline number, since it's the metric a residential customer actually asks about. A discounted cash-flow (DCF) view would be more rigorous for a large commercial project, but is unnecessary complexity for a small residential case; the 25-year cumulative view is a lightweight substitute that still credits degradation and one replacement cost, without a full year-by-year model.

Sensitivity Analysis

Base case: net investment Rs. 2,00,349 | tariff Rs. 8/kWh | generation 6,800 kWh/yr | self-consumption 55% | export rate Rs. 6.5/kWh | O&M Rs. 2,500/yr

1. Tariff Sensitivity - Payback (years)

Low (Rs. 6.5)	Base (Rs. 8)	High (Rs. 9.5)
4.8 yrs	4.2 yrs	3.8 yrs

2. Generation Sensitivity - Payback (years)

Lower (-10%)	Base	Higher (+10%)
4.7 yrs	4.2 yrs	3.8 yrs

3. Project Cost Sensitivity - Payback (years)

Lower (-10%)	Base	Higher (+10%)
3.8 yrs	4.2 yrs	4.7 yrs

Payback stays within a 3.8-4.8 year band across all three variables moving independently by 10% - the base case (~4.2 years) is not fragile to any single assumption.

Alternative Configurations Considered

Criterion	Selected: Waaree Bifacial + Fusion 63	Alt. 1 / Alt. 2
Module efficiency	22.26% (N-Type TOPCon, bifacial)	Alt.1: ~19-20% (Mono-PERC) Alt.2: ~22-23% (premium brand)
Approx. module cost	Rs. 11,299 (9 units)	Alt.1: ~15-20% cheaper/Wp Alt.2: ~10-15% pricier/Wp
Bifacial gain	Present, not counted in generation figure (conservative)	Alt.1: N/A Alt.2: depends on model
Warranty	12-yr product / 30-yr linear output	Alt.1: ~10-12yr/25yr Alt.2: similar to selected
Availability (India)	Good - domestic manufacturer, PM Surya Ghar-eligible	Both good; Alt.2 marginally higher lead time

Recommendation: the selected configuration is best balanced (bifacial, high efficiency, strong warranty) for this customer's budget-conscious priority. Alt. 1's lower cost comes with a shorter warranty and lower efficiency; Alt. 2's marginal efficiency gain doesn't justify its premium here.

The point of this comparison is not to declare one product universally 'best' - it's to show the trade-off reasoning: the selected configuration wins on this specific customer's stated priority, not on any single spec in isolation.